REPORT ON OPERATION RESOLVE & TRACE: SOC ANALYST CTF CHALLENGE

Course: Network Investigations & DNS Reconnaissance
Analyst: Horochi Chimezie
Target Domain: isc.org
Investigated IP: 204.152.184.88
Private Subnet: 172.31.1.0/24 — **excluded from findings because the scan did not complete**
Investigation Date: 10 August 2026

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1. Executive Summary

This investigation was conducted as a **CTF** exercise from a **SOC** perspective. The objective was to determine whether the investigated Internet-facing infrastructure could be associated with the (ISC), whether independent Internet intelligence supported that attribution, and what services were exposed.

DNS reconnaissance confirmed that **isc.org** has four IPv4 addresses, four IPv6 addresses, five authoritative name servers and two mail exchangers. The domain also uses Fastly for **www.isc.org**, showing that the domain and the investigated IP should not automatically be assumed to represent the same physical web service.

The investigated IP, **204.152.184.88**, did not currently return a PTR (Pointer) record during the local reverse-DNS query. The query returned `SERVFAIL`, so no current reverse-DNS name was recorded.

Independent network intelligence supplied for the investigation associates the surrounding network with **ISC** and **AS1280** . PeeringDB independently identifies AS1280 as Internet Systems Consortium.

A limited Nmap scan found only **TCP/80 (HTTP)** and **TCP/443 (HTTPS)** accessible. The remaining tested ports were filtered. Port 80 responded as a Go `net/http` server, while Nmap could not positively identify the HTTPS service version.

Overall, the evidence does **not** support treating 204.152.184.88 as malicious simply because it appeared in a SOC alert. The available evidence is more consistent with legitimate Internet-facing infrastructure. The main SOC concern is therefore to determine **why the internal host contacted this destination and whether that communication was expected**.

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2. Scope and Rules of Engagement

The investigation was limited to the supplied CTF (Capture The Flag) scope:

- isc.org
- 204.152.184.88
- Supplied CTF environment

The 172.31.1.0/24 internal network scan was attempted but could not be completed because Nmap returned **"Operation not permitted"** errors. Consequently, no conclusions about that subnet are included in this report.

Although 172.31.1.50 was reported as up, the subsequent service scan also encountered packet-transmission errors and returned all scanned ports as filtered. These results are therefore not sufficient to establish which hosts or services are actually exposed on the internal network.

Reconnaissance was limited to the minimum necessary DNS, OSINT , connectivity and service-enumeration activities required for the assignment.

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3. DNS Reconnaissance

DNS Record Matrix

Record	Command	Result
A	`dig isc.org A +short`	`151.101.2.217` `151.101.66.217` `151.101.130.217` `151.101.194.217`
AAAA	`dig isc.org AAAA +short`	`2a04:4e42::729` `2a04:4e42:200::729` `2a04:4e42:400::729` `2a04:4e42:600::729`
MX	`dig isc.org MX +short`	`5 mx.pao1.isc.org.` `10 mx.ams1.isc.org.`

NS	`dig isc.org NS +short`	ns1.isc.org. ns2.isc.org. ns3.isc.org. ns.isc.afilias-nst.info. nsp.dnsnode.net.
CNAME	`dig isc.org CNAME`	No CNAME returned
PTR	`dig -x 204.152.184.88 +short`	**No result; reverse query returned SERVFAIL**

DNS interpretation

The **A (Address)** records identify the current IPv4 addresses returned for `isc.org`. They are all in the `151.101.x.x` range.

The **AAAA (IPv6 Address)** records provide IPv6 connectivity for the domain.

The **MX (Mail Exchange)** records identify the servers responsible for receiving mail for `isc.org`. The lower preference value, 5, makes `mx.pao1.isc.org` the preferred exchanger over `mx.ams1.isc.org`.

The **NS (Name Server)** records identify the authoritative DNS servers for the domain.

The direct query for a CNAME (Canonical Name) on `isc.org` returned no CNAME. However, `www.isc.org` does use:

`isc.map.fastlydns.net`.

which resolves to:

`199.232.58.217`

This indicates that ISC's web presence uses external/CDN (Content Delivery Network) infrastructure for at least the www hostname.

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4. DNSSEC and Trace Observations

The `dig isc.org +trace` investigation successfully followed the DNS delegation from the root zone to `.org` and then to the authoritative ISC name servers.

The trace showed:

```
Root (.)
↓
.org
↓
isc.org
↓
ns1.isc.org / ns2.isc.org / ns3.isc.org / other authoritative servers
```

The trace also returned **DS (Delegation Signer)** and **RRSIG (Resource Record Signature)** records.

These are indicators of **DNSSEC (Domain Name System Security Extensions)** being used to protect DNS data integrity.

The trace eventually returned the same four IPv4 addresses for `isc.org`:

- 151.101.2.217
- 151.101.66.217
- 151.101.130.217
- 151.101.194.217

The trace encountered an IPv6 routing limitation:

`network unreachable`

when attempting to contact one of the authoritative servers over IPv6. This is a limitation of the investigation environment rather than evidence of malicious DNS activity.

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5. Reverse DNS

The investigated address was:

204.152.184.88

The reverse DNS name queried was:

`88.184.152.204.in-addr.arpa`

The current query returned:

`SERVFAIL`

and the `+short` query returned no hostname.

Therefore:

Current PTR finding: No PTR result observed from the investigation environment.

This is important because a missing or failed PTR lookup does not, by itself, indicate malicious activity.

Historical information supplied as part of the investigation associated this address with `external.isc.org`. However, that historical information is **not treated as the current PTR result**.

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6. Internet Intelligence and Ownership

Attribute	Finding
Investigated IP	**204.152.184.88**
Organization	**Internet Systems Consortium (ISC)**
ASN (Autonomous System Number)	**AS1280**
Country	**United States**
Network context	ISC-associated infrastructure
Current PTR	No result / SERVFAIL
BGP (Border Gateway Protocol) context	Consistent with ISC network attribution
WHOIS/RIR	Supports Internet-number resource attribution
VirusTotal	Investigated; no specific detection count recorded in supplied evidence
AbuseIPDB	Investigated; no specific score recorded in supplied evidence

Independent network information identifies **AS1280** with Internet Systems Consortium. PeeringDB describes AS1280 as the network operated by ISC and notes that ISC operates the F-Root nameserver infrastructure.

This is important from a SOC perspective because the ownership evidence is independent of the DNS records. It provides another layer of evidence supporting legitimate organizational attribution.

OSINT resources consulted

- VirusTotal IP investigation
- AbuseIPDB IP investigation
- IP2Location lookup
- IPLocation lookup
- Hurricane Electric BGP Toolkit
- DNSChecker DNS lookup

VirusTotal's documentation confirms that IP intelligence can include ASN, geographic information, WHOIS and passive-DNS relationships.

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7. Attack Surface — Public IP

Nmap Results

Port	State	Service	Version / Observation
80/tcp	**Open**	HTTP	Go `net/http` server
443/tcp	**Open**	HTTPS	Service detected, version not positively identified
22/tcp	Filtered	SSH	No response
25/tcp	Filtered	SMTP	No response
53/tcp	Filtered	DNS	No response

110/tcp	Filtered	POP3	No response
143/tcp	Filtered	IMAP	No response
587/tcp	Filtered	SMTP Submission	No response
993/tcp	Filtered	IMAPS	No response
995/tcp	Filtered	POP3S	No response

The general Nmap scan showed **998 of the 1000 scanned TCP ports as filtered**, with only ports 80 and 443 open.

The nc (Netcat) connectivity checks independently confirmed:

- `204.152.184.88:80` — **open**
- `204.152.184.88:443` — **open**

Service assessment

HTTP and HTTPS are reasonable services to find on Internet-facing organizational infrastructure.

Therefore, the presence of ports 80 and 443 is **not itself evidence of compromise**.

Nmap identified port 80 as a Go `net/http` server. It could not confidently identify the HTTPS service version, so no vulnerability claim is made from the scan alone.

No evidence was collected showing that the exposed services were outdated or vulnerable.

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8. SOC Threat Assessment

Finding	Assessment
DNS ownership	**Low concern**

ASN/network ownership	**Low concern**
Forward DNS	**Consistent with ISC domain infrastructure**
Reverse DNS	**Unconfirmed; current query returned SERVFAIL**
Reputation intelligence	**No malicious verdict established from supplied evidence**
HTTP/HTTPS exposure	**Expected for public web infrastructure**
Vulnerable service	**Not established**
Suspicious DNS relationship	**Not established**

Risk Score

The following scoring model was supplied for the CTF (Capture The Flag) exercise. It is an **analytical model, not an industry-standard risk framework**.

Finding	Score
Legitimate organizational ownership	**-2**
Consistent forward/reverse DNS	**0**
Clean/low reputation	**0**
Known malicious reports	**0**
Unexpected open services	**0**
Vulnerable/outdated service	**0**
Suspicious DNS relationships	**0**
Total	**-2**

Classification: LOW

The negative score is retained because the supplied scoring model awards a negative value for legitimate ownership. Operationally, the result falls within the model's **Low** category.

The open web services were not scored as "unexpected" because HTTP and HTTPS are normal services for a public-facing organization.

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9. SOC Interpretation

The most important conclusion is that **an alert involving an IP address does not automatically mean that the IP is malicious**.

In this investigation, several independent pieces of evidence point toward legitimate ISC infrastructure:

1. The target domain is clearly associated with ISC.
2. The DNS infrastructure contains multiple ISC-operated nameservers.
3. Independent network intelligence associates AS1280 with Internet Systems Consortium.
4. The investigated IP is associated with the ISC network context supplied for this exercise.
5. The exposed services are limited primarily to HTTP and HTTPS.
6. No vulnerable or clearly malicious service was identified.

There is also an important distinction between the domain's current A records and the investigated IP. The current `isc.org` A records returned `151.101.x.x` addresses, not `204.152.184.88`. Therefore, the investigation should **not claim that 204.152.184.88 is currently the direct A record for isc.org**.

Instead, the evidence supports the more careful conclusion that the investigated IP is associated with legitimate ISC network infrastructure.

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10. Recommendations

From a SOC (Security Operations Center) perspective, the next step should be to investigate the **reason for the communication**, rather than blocking the destination solely because an alert was generated.

1. Identify the internal source

Determine:

- Which internal workstation/server contacted `204.152.184.88`?
- When did the connection occur?
- Which process generated it?
- Was the connection initiated by a browser, system service or application?

2. Review DNS logs

Look for:

- Repeated queries
- Unusual subdomains
- High query frequency
- NXDOMAIN (Non-Existent Domain) spikes
- Random-looking domain names
- DNS activity inconsistent with the host's normal role

3. Review network traffic

Correlate:

- Source IP
- Destination IP
- Destination port
- Timestamp
- Protocol
- Bytes sent/received
- Number and frequency of connections

4. Validate business purpose

If the internal system legitimately accesses ISC services, document the business reason and consider the alert a potential false positive.

5. Continue monitoring

If the communication is unexpected, investigate the source host for additional indicators rather than treating the destination IP alone as evidence of compromise.

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11. Final Conclusion

Based on the evidence collected during this investigation, **204.152.184.88 should not currently be classified as malicious**.

The strongest evidence points toward legitimate ISC-associated infrastructure. DNS reconnaissance confirms legitimate ISC domain infrastructure, while independent network intelligence associates AS1280 with Internet Systems Consortium. The public IP exposes HTTP and HTTPS, but no obviously unnecessary or vulnerable service was identified.

The current reverse-DNS query failed with `SERVFAIL`, so the investigation does not claim a current PTR record for the IP.

The most appropriate SOC conclusion is therefore:

> **The investigated public IP is attributable to legitimate ISC-associated infrastructure. DNS and network ownership evidence are broadly consistent, while the available reputation and service evidence does not independently establish malicious activity. Any SIEM (Security Information and Event Management) alert involving this destination should therefore be investigated in the context of the specific internal source host, destination port, DNS activity, timing and connection behaviour rather than treating the destination IP alone as evidence of compromise.**

Overall CTF analytical risk: LOW.

Abbreviations Used

Abbreviation	Meaning
CTF	Capture The Flag
SOC	Security Operations Center
SIEM	Security Information and Event Management
DNS	Domain Name System
DNSSEC	Domain Name System Security Extensions

IP	Internet Protocol
A	IPv4 Address record
AAAA	IPv6 Address record
MX	Mail Exchange
NS	Name Server
CNAME	Canonical Name
PTR	Pointer / Reverse-DNS record
ASN	Autonomous System Number
BGP	Border Gateway Protocol
WHOIS	Internet registration/ownership lookup service
RIR	Regional Internet Registry
OSINT	Open-Source Intelligence
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
Nmap	Network Mapper
NC / Netcat	Network connection and diagnostic utility
VT	VirusTotal
ROA	Route Origin Authorization
CDN	Content Delivery Network

RRSIG	Resource Record Signature
DS	Delegation Signer
SERVFAIL	Server Failure DNS response

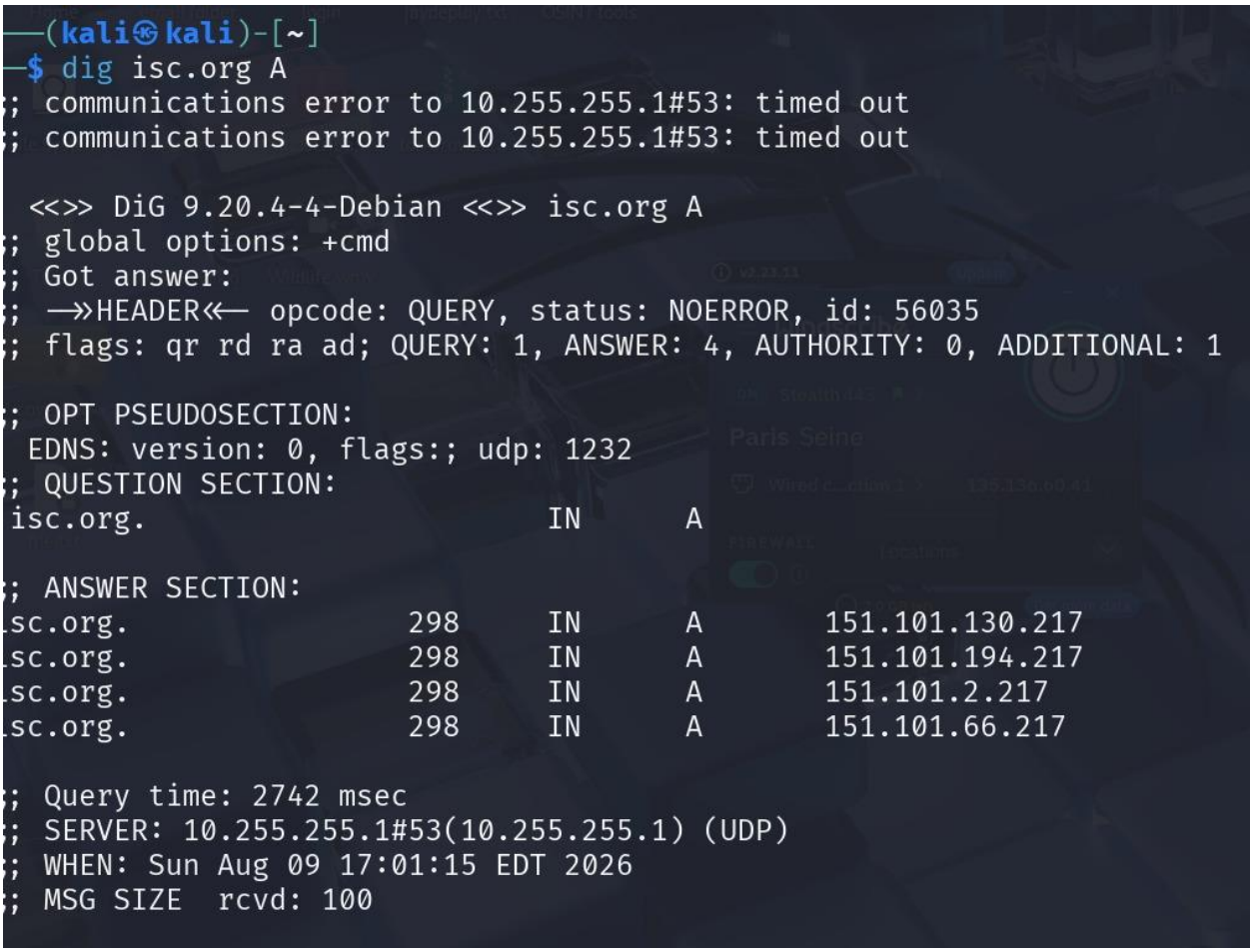
BELOW ARE THE SCREENSHOT OF THE SCANS ON KALI LINUX

```
┌──(kali㉿kali)-[~]
└─$ whoami
kali

┌──(kali㉿kali)-[~]
└─$ pwd
/home/kali

┌──(kali㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:42:0a:f3 brd ff:ff:ff:ff:ff:ff
    inet 10.18.3.116/24 brd 10.18.3.255 scope global dynamic noprefixroute eth0
       valid_lft 176sec preferred_lft 176sec
    inet6 fe80::40f2:3bac:9e61:59e1/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
10: tun0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 500
    link/none
    inet 10.145.204.141/22 brd 10.145.207.255 scope global tun0
       valid_lft forever preferred_lft forever
    inet6 fe80::1f74:88d6:1a55:b532/64 scope link stable-privacy proto kernel_ll
       valid_lft forever preferred_lft forever
```

```
┌──(kali㉿kali)-[~]
└─$ ip r
0.0.0.0/1 via 10.145.204.1 dev tun0
default via 10.18.3.35 dev eth0 proto dhcp src 10.18.3.116 metric 100
10.18.3.0/24 dev eth0 proto kernel scope link src 10.18.3.116 metric 100
10.145.204.0/22 dev tun0 proto kernel scope link src 10.145.204.141
10.255.255.0/24 dev tun0 scope link
127.0.0.1 via 10.18.3.35 dev eth0
128.0.0.0/1 via 10.145.204.1 dev tun0
135.136.60.36 via 10.18.3.35 dev eth0
```

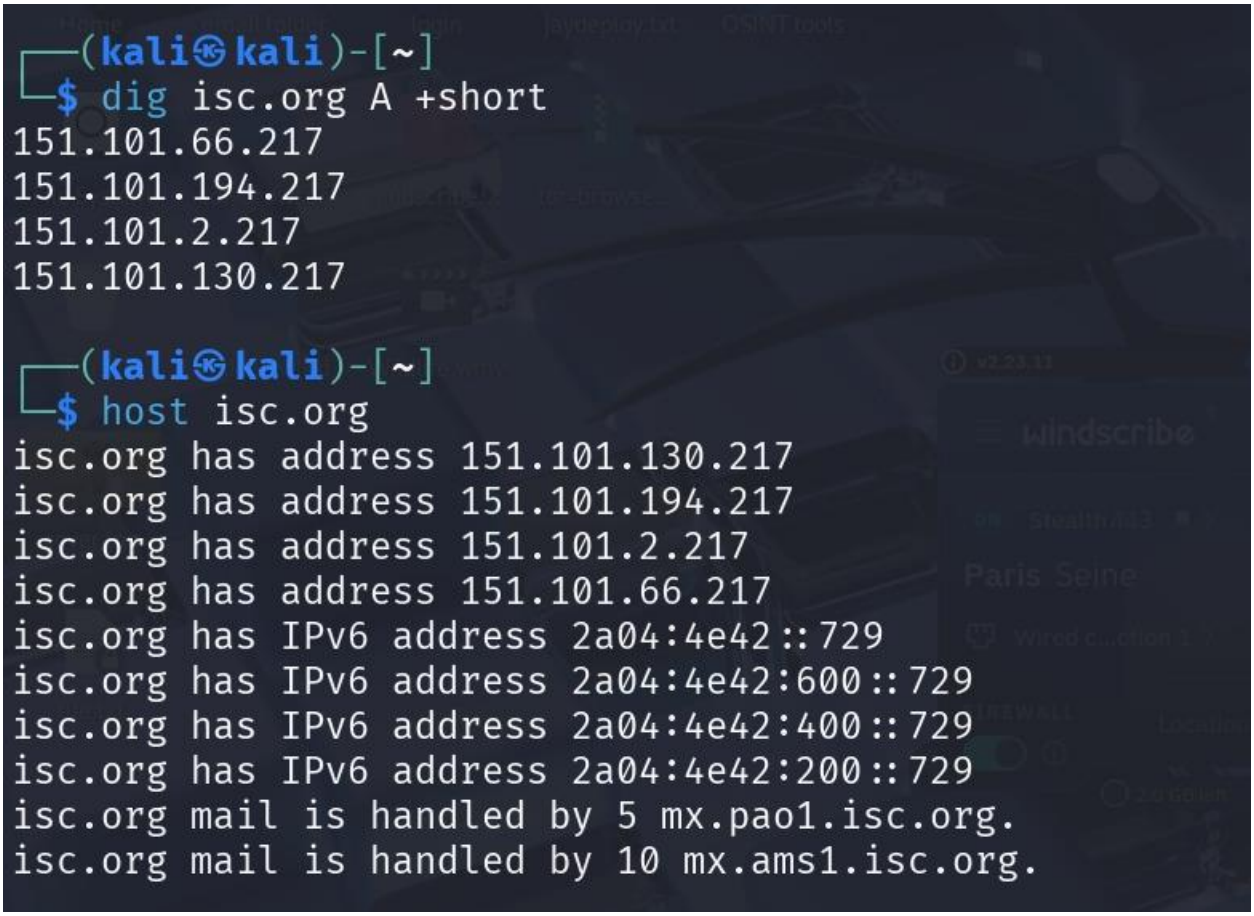

┌──(kali㉿kali)-[~]
└─$ dig isc.org A
; communications error to 10.255.255.1#53: timed out
; communications error to 10.255.255.1#53: timed out

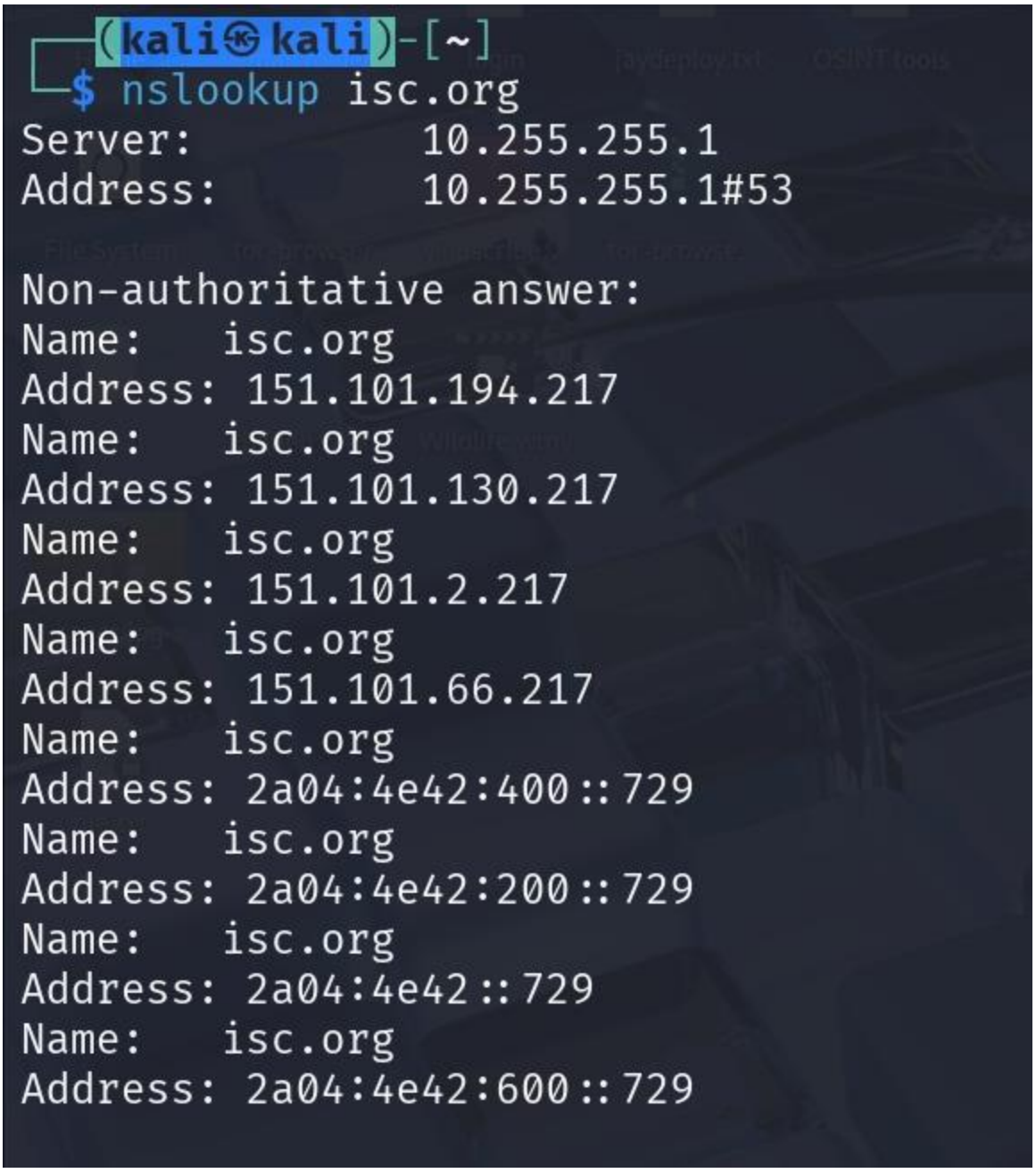
<<>> DiG 9.20.4-4-Debian <<>> isc.org A
; global options: +cmd
; Got answer:
; ⟶»HEADER«⟵ opcode: QUERY, status: NOERROR, id: 56035
; flags: qr rd ra ad; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 1

; OPT PSEUDOSECTION:
EDNS: version: 0, flags:; udp: 1232
; QUESTION SECTION:
isc.org. IN A

; ANSWER SECTION:
isc.org. 298 IN A 151.101.130.217
isc.org. 298 IN A 151.101.194.217
isc.org. 298 IN A 151.101.2.217
isc.org. 298 IN A 151.101.66.217

; Query time: 2742 msec
; SERVER: 10.255.255.1#53(10.255.255.1) (UDP)
; WHEN: Sun Aug 09 17:01:15 EDT 2026
; MSG SIZE rcvd: 100


┌──(kali㉿kali)-[~]
└─$ dig isc.org A +short
151.101.66.217
151.101.194.217
151.101.2.217
151.101.130.217
┌──(kali㉿kali)-[~]
└─$ host isc.org
isc.org has address 151.101.130.217
isc.org has address 151.101.194.217
isc.org has address 151.101.2.217
isc.org has address 151.101.66.217
isc.org has IPv6 address 2a04:4e42::729
isc.org has IPv6 address 2a04:4e42:600::729
isc.org has IPv6 address 2a04:4e42:400::729
isc.org has IPv6 address 2a04:4e42:200::729
isc.org mail is handled by 5 mx.pao1.isc.org.
isc.org mail is handled by 10 mx.ams1.isc.org.


(kali㉿kali)-[~]
$ nslookup isc.org
Server: 10.255.255.1
Address: 10.255.255.1#53
Non-authoritative answer:
Name: isc.org
Address: 151.101.194.217
Name: isc.org
Address: 151.101.130.217
Name: isc.org
Address: 151.101.2.217
Name: isc.org
Address: 151.101.66.217
Name: isc.org
Address: 2a04:4e42:400::729
Name: isc.org
Address: 2a04:4e42:200::729
Name: isc.org
Address: 2a04:4e42::729
Name: isc.org
Address: 2a04:4e42:600::729

```
┌──(kali㉿kali)-[~]
└─$ isc.org → IPv4 address
isc.org: command not found

┌──(kali㉿kali)-[~]
└─$ dig www.isc.org A +short
isc.map.fastlydns.net.
199.232.54.217

┌──(kali㉿kali)-[~]
└─$ dig isc.org AAAA

; <<>> DiG 9.20.4-4-Debian <<>> isc.org AAAA
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 32052
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;isc.org.                       IN      AAAA

;; ANSWER SECTION:
isc.org.                36      IN      AAAA    2a04:4e42::729
isc.org.                36      IN      AAAA    2a04:4e42:400::729
isc.org.                36      IN      AAAA    2a04:4e42:200::729
isc.org.                36      IN      AAAA    2a04:4e42:600::729

;; Query time: 868 msec
;; SERVER: 10.255.255.1#53(10.255.255.1) (UDP)
;; WHEN: Sun Aug 09 17:07:39 EDT 2026
;; MSG SIZE  rcvd: 148
```

```
┌──(kali㉿kali)-[~]
└─$ dig isc.org AAAA +short
2a04:4e42:200::729
2a04:4e42:400::729
2a04:4e42:600::729
2a04:4e42::729

┌──(kali㉿kali)-[~]
└─$ dig isc.org MX

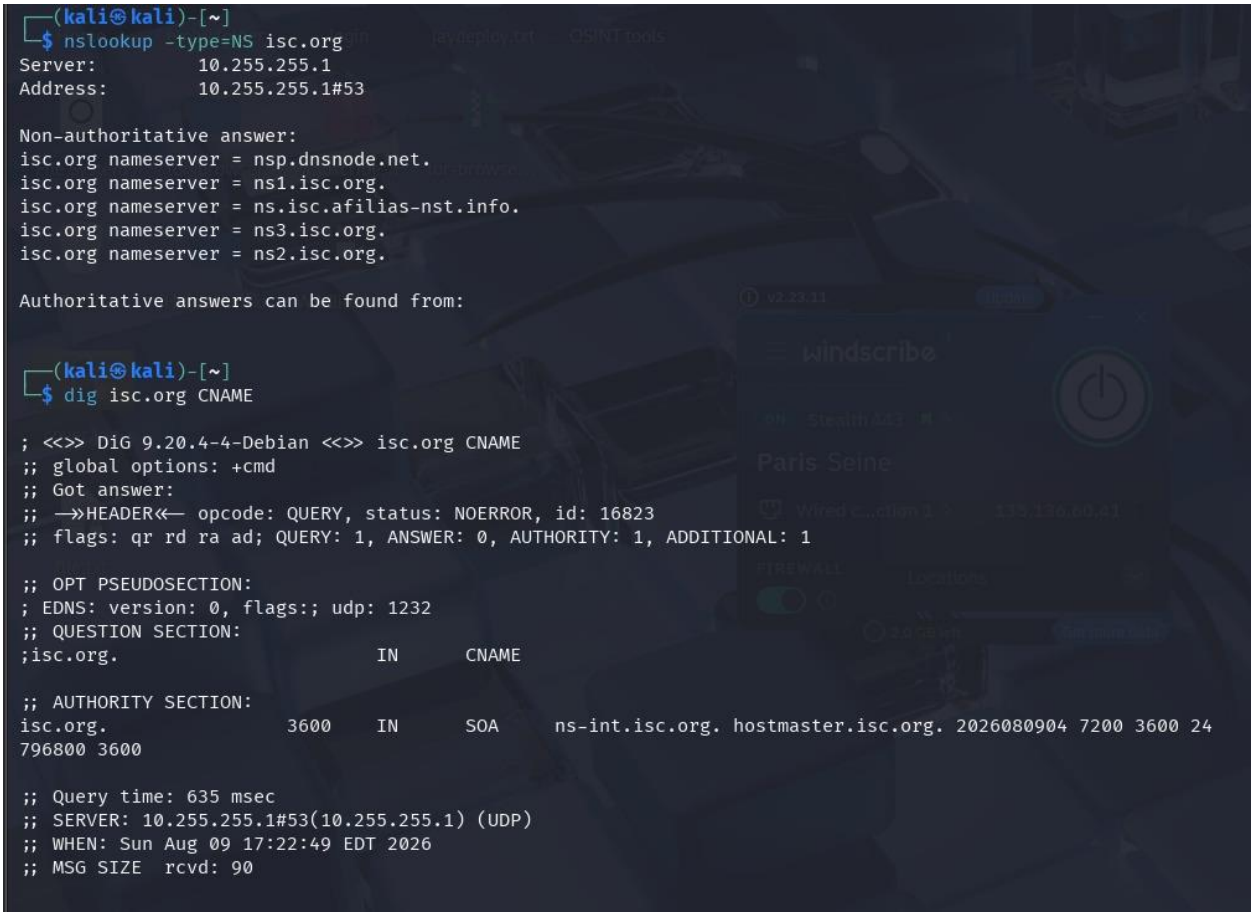
; <<>> DiG 9.20.4-4-Debian <<>> isc.org MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 43218
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;isc.org.                       IN      MX

;; ANSWER SECTION:
isc.org.                300     IN      MX      5 mx.pao1.isc.org.
isc.org.                300     IN      MX      10 mx.ams1.isc.org.

;; Query time: 4157 msec
;; SERVER: 10.255.255.1#53(10.255.255.1) (UDP)
;; WHEN: Sun Aug 09 17:14:50 EDT 2026
;; MSG SIZE  rcvd: 84


┌──(kali㉿kali)-[~]
└─$ dig isc.org MX +short
5 mx.pao1.isc.org.
10 mx.ams1.isc.org.
```



```
┌──(kali㉿kali)-[~]
└─$ nslookup -type=NS isc.org
Server:         10.255.255.1
Address:        10.255.255.1#53

Non-authoritative answer:
isc.org nameserver = nsp.dnsnode.net.
isc.org nameserver = ns1.isc.org.
isc.org nameserver = ns.isc.afilias-nst.info.
isc.org nameserver = ns3.isc.org.
isc.org nameserver = ns2.isc.org.

Authoritative answers can be found from:

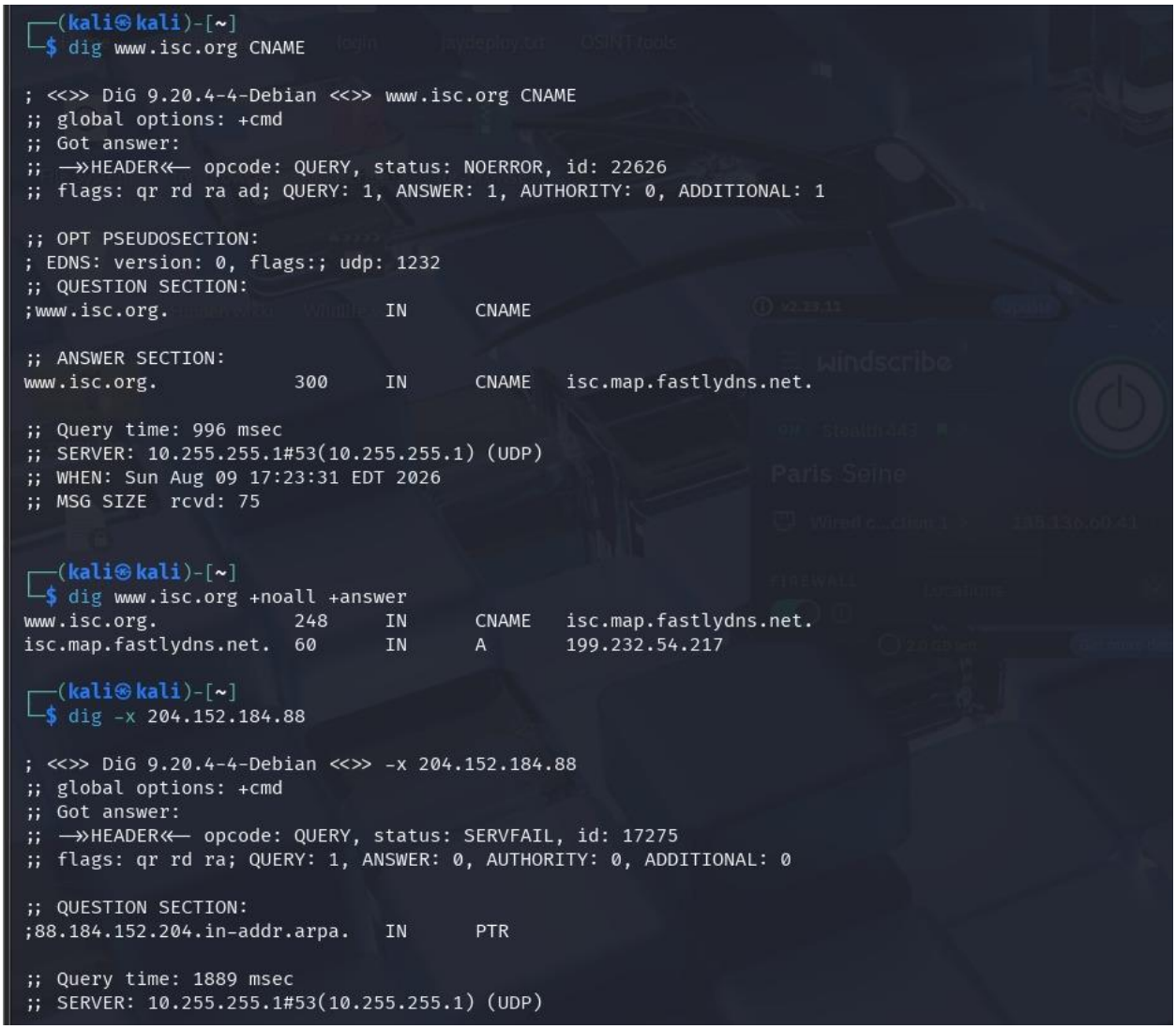

┌──(kali㉿kali)-[~]
└─$ dig isc.org CNAME

; <<>> DiG 9.20.4-4-Debian <<>> isc.org CNAME
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 16823
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;isc.org.                       IN      CNAME

;; AUTHORITY SECTION:
isc.org.                3600    IN      SOA     ns-int.isc.org. hostmaster.isc.org. 2026080904 7200 3600 24
796800 3600

;; Query time: 635 msec
;; SERVER: 10.255.255.1#53(10.255.255.1) (UDP)
;; WHEN: Sun Aug 09 17:22:49 EDT 2026
;; MSG SIZE  rcvd: 90
```



```
┌──(kali㉿kali)-[~]
└─$ dig www.isc.org CNAME

; <<>> DiG 9.20.4-4-Debian <<>> www.isc.org CNAME
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 22626
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;www.isc.org.                   IN      CNAME

;; ANSWER SECTION:
www.isc.org.            300     IN      CNAME   isc.map.fastlydns.net.

;; Query time: 996 msec
;; SERVER: 10.255.255.1#53(10.255.255.1) (UDP)
;; WHEN: Sun Aug 09 17:23:31 EDT 2026
;; MSG SIZE  rcvd: 75

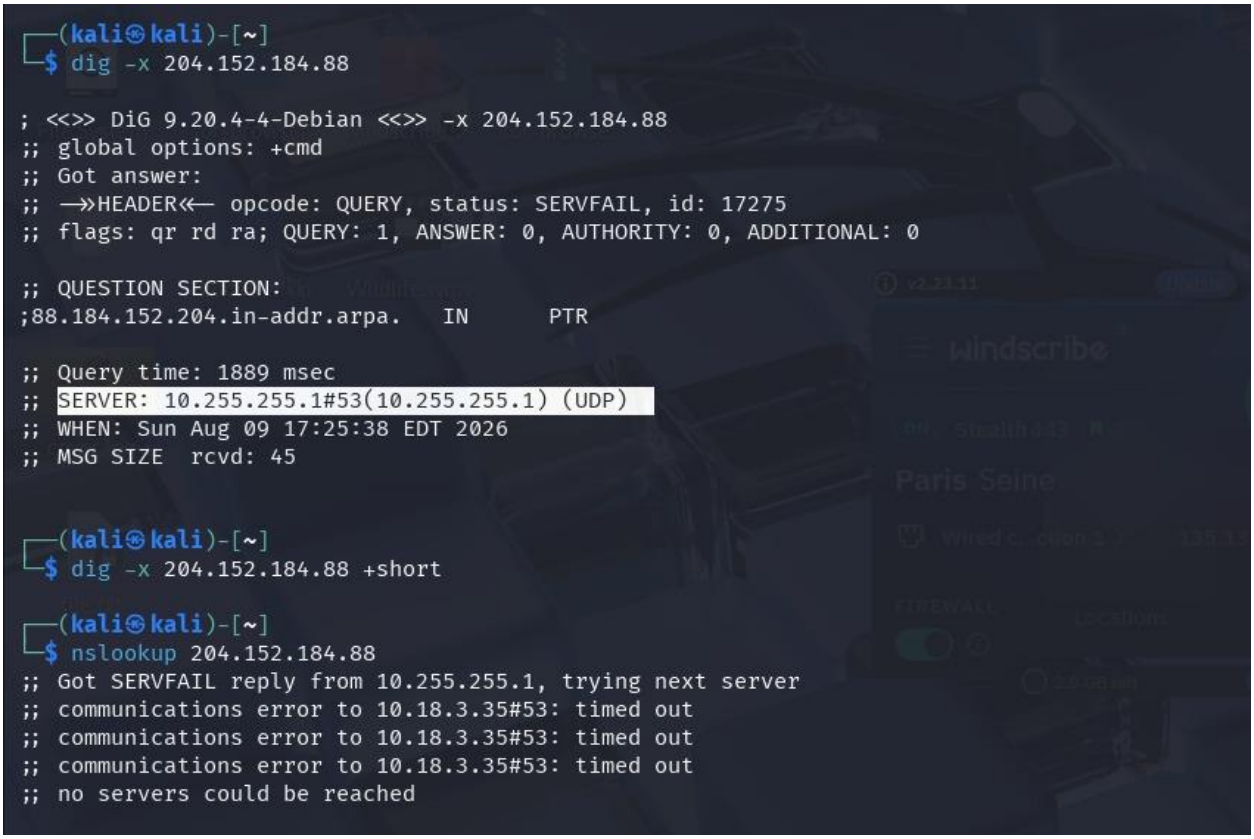

┌──(kali㉿kali)-[~]
└─$ dig www.isc.org +noall +answer
www.isc.org.            248     IN      CNAME   isc.map.fastlydns.net.
isc.map.fastlydns.net.  60      IN      A       199.232.54.217

┌──(kali㉿kali)-[~]
└─$ dig -x 204.152.184.88

; <<>> DiG 9.20.4-4-Debian <<>> -x 204.152.184.88
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 17275
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;88.184.152.204.in-addr.arpa.   IN      PTR

;; Query time: 1889 msec
;; SERVER: 10.255.255.1#53(10.255.255.1) (UDP)
```



```
┌──(kali㉿kali)-[~]
└─$ dig -x 204.152.184.88

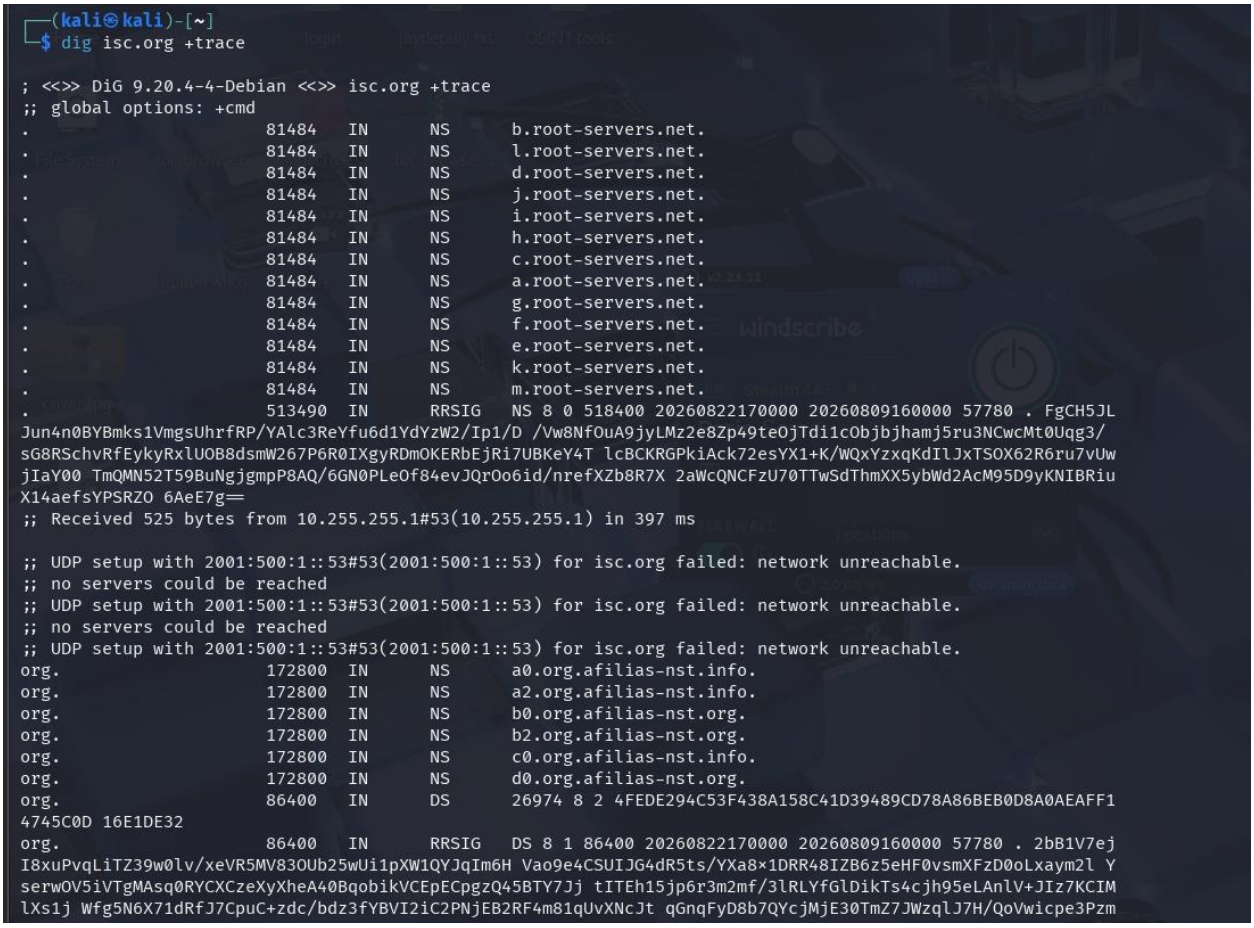
; <<>> DiG 9.20.4-4-Debian <<>> -x 204.152.184.88
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 17275
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;88.184.152.204.in-addr.arpa.   IN      PTR

;; Query time: 1889 msec
;; SERVER: 10.255.255.1#53(10.255.255.1) (UDP)
;; WHEN: Sun Aug 09 17:25:38 EDT 2026
;; MSG SIZE  rcvd: 45


┌──(kali㉿kali)-[~]
└─$ dig -x 204.152.184.88 +short

┌──(kali㉿kali)-[~]
└─$ nslookup 204.152.184.88
;; Got SERVFAIL reply from 10.255.255.1, trying next server
;; communications error to 10.18.3.35#53: timed out
;; communications error to 10.18.3.35#53: timed out
;; communications error to 10.18.3.35#53: timed out
;; no servers could be reached
```



```
┌──(kali㉿kali)-[~]
└─$ dig isc.org +trace

; <<>> DiG 9.20.4-4-Debian <<>> isc.org +trace
;; global options: +cmd
.                       81484   IN      NS      b.root-servers.net.
.                       81484   IN      NS      l.root-servers.net.
.                       81484   IN      NS      d.root-servers.net.
.                       81484   IN      NS      j.root-servers.net.
.                       81484   IN      NS      i.root-servers.net.
.                       81484   IN      NS      h.root-servers.net.
.                       81484   IN      NS      c.root-servers.net.
.                       81484   IN      NS      a.root-servers.net.
.                       81484   IN      NS      g.root-servers.net.
.                       81484   IN      NS      f.root-servers.net.
.                       81484   IN      NS      e.root-servers.net.
.                       81484   IN      NS      k.root-servers.net.
.                       81484   IN      NS      m.root-servers.net.
.                       513490  IN      RRSIG   NS 8 0 518400 20260822170000 20260809160000 57780 . FgCH5JL
Jun4n0BYBmks1VmgsUhrfRP/YAlc3ReYfu6d1YdYzW2/Ip1/D /Vw8NfOuA9jyLMz2e8Zp49teOjTdi1cObjbjhamj5ru3NCwcMt0Uqg3/
sG8RSchvRfEykyRxlUOB8dsmW267P6R0IXgyRDmOKERbEjRi7UBKeY4T lcBCKRGPkiAck72esYX1+K/WQxYzxqKdIlJxTSOX62R6ru7vUw
jIaY00 TmQMN52T59BuNgjgmpP8AQ/6GN0PLeOf84evJQrOo6id/nrefXZb8R7X 2aWcQNCFzU70TTwSdThmXX5ybWd2AcM95D9yKNIBRiu
X14aefsYPSRZO 6AeE7g==
;; Received 525 bytes from 10.255.255.1#53(10.255.255.1) in 397 ms

;; UDP setup with 2001:500:1::53#53(2001:500:1::53) for isc.org failed: network unreachable.
;; no servers could be reached
;; UDP setup with 2001:500:1::53#53(2001:500:1::53) for isc.org failed: network unreachable.
;; no servers could be reached
;; UDP setup with 2001:500:1::53#53(2001:500:1::53) for isc.org failed: network unreachable.
org.                    172800  IN      NS      a0.org.afilias-nst.info.
org.                    172800  IN      NS      a2.org.afilias-nst.info.
org.                    172800  IN      NS      b0.org.afilias-nst.org.
org.                    172800  IN      NS      b2.org.afilias-nst.org.
org.                    172800  IN      NS      c0.org.afilias-nst.info.
org.                    172800  IN      NS      d0.org.afilias-nst.org.
org.                    86400   IN      DS      26974 8 2 4FEDE294C53F438A158C41D39489CD78A86BEB0D8A0AEAFF1
4745C0D 16E1DE32
org.                    86400   IN      RRSIG   DS 8 1 86400 20260822170000 20260809160000 57780 . 2bB1V7ej
I8xuPvqLiTZ39w0lv/xeVR5MV83OUb25wUi1pXW1QYJqIm6H Vao9e4CSUIJG4dR5ts/YXa8×1DRR48IZB6z5eHF0vsmXFzD0oLxaym2l Y
serwOV5iVTgMAsq0RYCXCzeXyXheA40BqobikVCEpECpgzQ45BTY7Jj tITEh15jp6r3m2mf/3lRLYfGlDikTs4cjh95eLAnlV+JIz7KCIM
lXs1j Wfg5N6X71dRfJ7CpuC+zdc/bdz3fYBVI2iC2PNjEB2RF4m81qUvXNcJt qGnqFyD8b7QYcjMjE30TmZ7JWzqlJ7H/QoVwicpe3Pzm
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I8xuPvqLiTZ39w0lv/xeVR5MV83OUb25wUi1pXW1QYJqIm6H Vao9e4CSUIJG4dR5ts/YXa8×1DRR48IZB6z5eHF0vsmXFzD0oLxaym2l Y
serwOV5iVTgMAsq0RYCXCzeXyXheA40BqobikVCEpECpgzQ45BTY7Jj tITEh15jp6r3m2mf/3lRLYfGlDikTs4cjh95eLAnlV+JIz7KCIM
lXs1j Wfg5N6X71dRfJ7CpuC+zdc/bdz3fYBVI2iC2PNjEB2RF4m81qUvXNcJt qGnqFyD8b7QYcjMjE30TmZ7JWzqlJ7H/QoVwicpe3Pzm
ms/lgEZXRjQk PzQkCw==
;; Received 773 bytes from 192.58.128.30#53(j.root-servers.net) in 622 ms

;; UDP setup with 2001:500:48::1#53(2001:500:48::1) for isc.org failed: network unreachable.
isc.org.                3600    IN      NS      ns.isc.afilias-nst.info.
isc.org.                3600    IN      NS      nsp.dnsnode.net.
isc.org.                3600    IN      NS      ns1.isc.org.
isc.org.                3600    IN      NS      ns2.isc.org.
isc.org.                3600    IN      NS      ns3.isc.org.
isc.org.                3600    IN      DS      7250 13 2 A30B3F78B6DDE9A4A9A2AD0C805518B4F49EC62E7D3F4531D
33DE697 CDA01CB2
isc.org.                3600    IN      RRSIG   DS 8 2 3600 20260830152406 20260809142406 44925 org. MzoQwP
nbm0bGfirNJr1asrqdI2pGhsOra87cHH9ztWeDeOOFsMmLRNzM PTBYPlOJ/8/vLCvos8IyDLMqdIslDeoF1tDfPeN5V94pvBqGd7PpS0fb
 cFmrtBxyPMoj5btAg33CBuS5ifExzVFiZJfWqT7qEX8NsrvrEBrulaA3 tvo=
;; Received 499 bytes from 199.19.56.1#53(a0.org.afilias-nst.info) in 405 ms

;; UDP setup with 2001:500:6b:2::26#53(2001:500:6b:2::26) for isc.org failed: network unreachable.
isc.org.                300     IN      A       151.101.2.217
isc.org.                300     IN      A       151.101.194.217
isc.org.                300     IN      A       151.101.130.217
isc.org.                300     IN      A       151.101.66.217
isc.org.                300     IN      RRSIG   A 13 2 300 20260815184324 20260801175109 27566 isc.org. +QC
gbAqAx+TJo3VBNo9pOeR4PqOinr4NOzsvEyiOdqJU/xlxPdH5MopP 8g66Oz3nAa1SngNc1TYgFCPkRDtHog==
isc.org.                7200    IN      NS      ns1.isc.org.
isc.org.                7200    IN      NS      ns3.isc.org.
isc.org.                7200    IN      NS      nsp.dnsnode.net.
isc.org.                7200    IN      NS      ns.isc.afilias-nst.info.
isc.org.                7200    IN      NS      ns2.isc.org.
isc.org.                7200    IN      RRSIG   NS 13 2 7200 20260815184324 20260801175109 27566 isc.org. W
ptAd7svbknoDuoIfeyaVpWwWobY4RBMiED80nJNqf70vnXQl2Twr6my OM0xlAJBXnBzGed3tiq+Rku/uzt8PA==
;; Received 1204 bytes from 51.75.79.143#53(ns3.isc.org) in 373 ms


┌──(kali㉿kali)-[~]
└─$
```